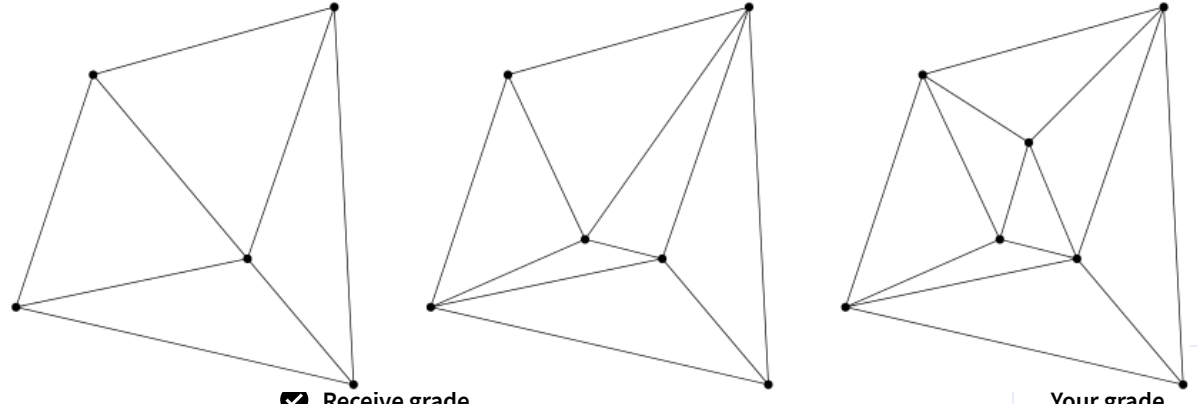


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1.



1 / 1 point

Receive grade

Your grade

100%

The image above shows two steps of the randomized incremental construction algorithm. How many triangles were done in total during the insertion of these two points?

2

Correct

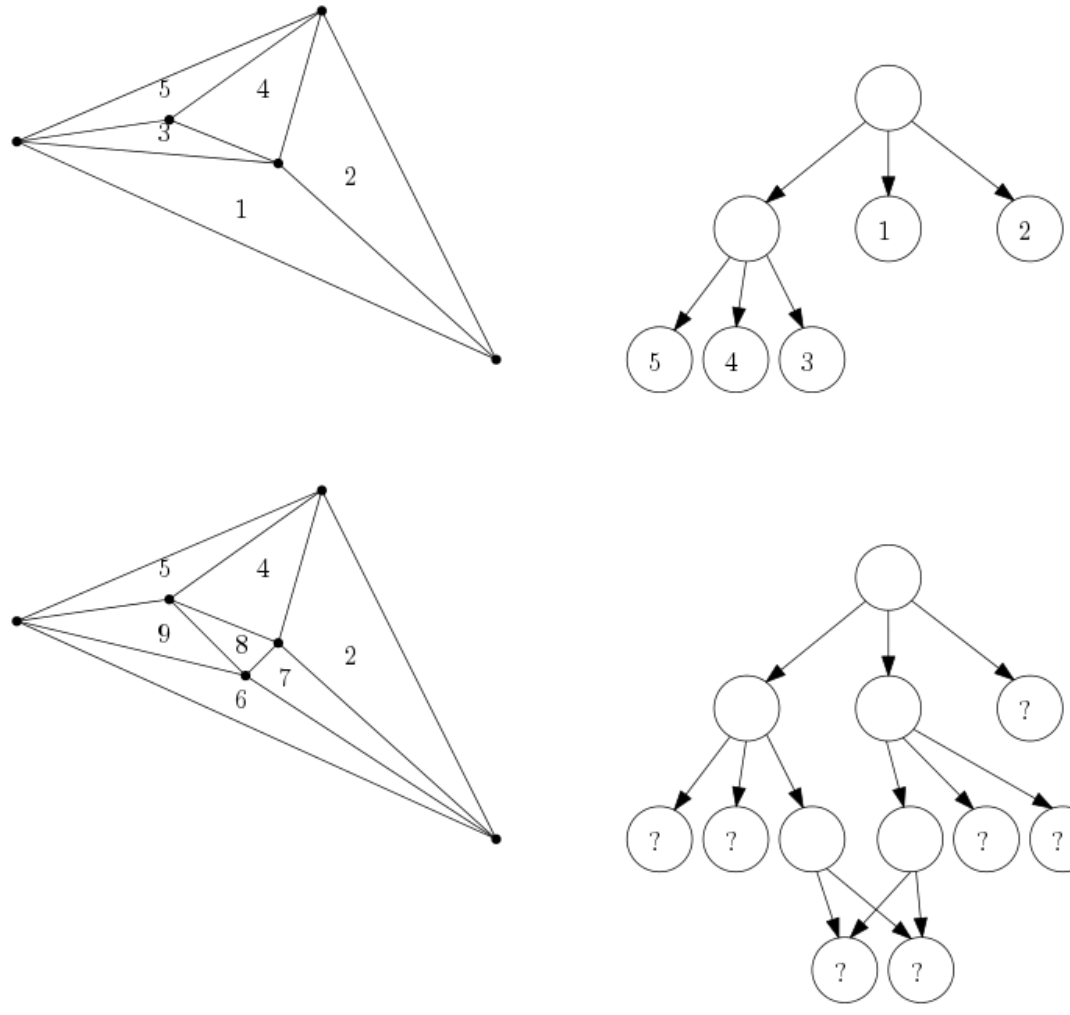
Correct

Like

Dislike

Report an issue

2.



1 / 1 point

The image above shows one step of the randomized incremental construction together with their point location structures. What is the correct labelling from left to right of the data structure at the bottom right?

5 4 8 9 6 7 2

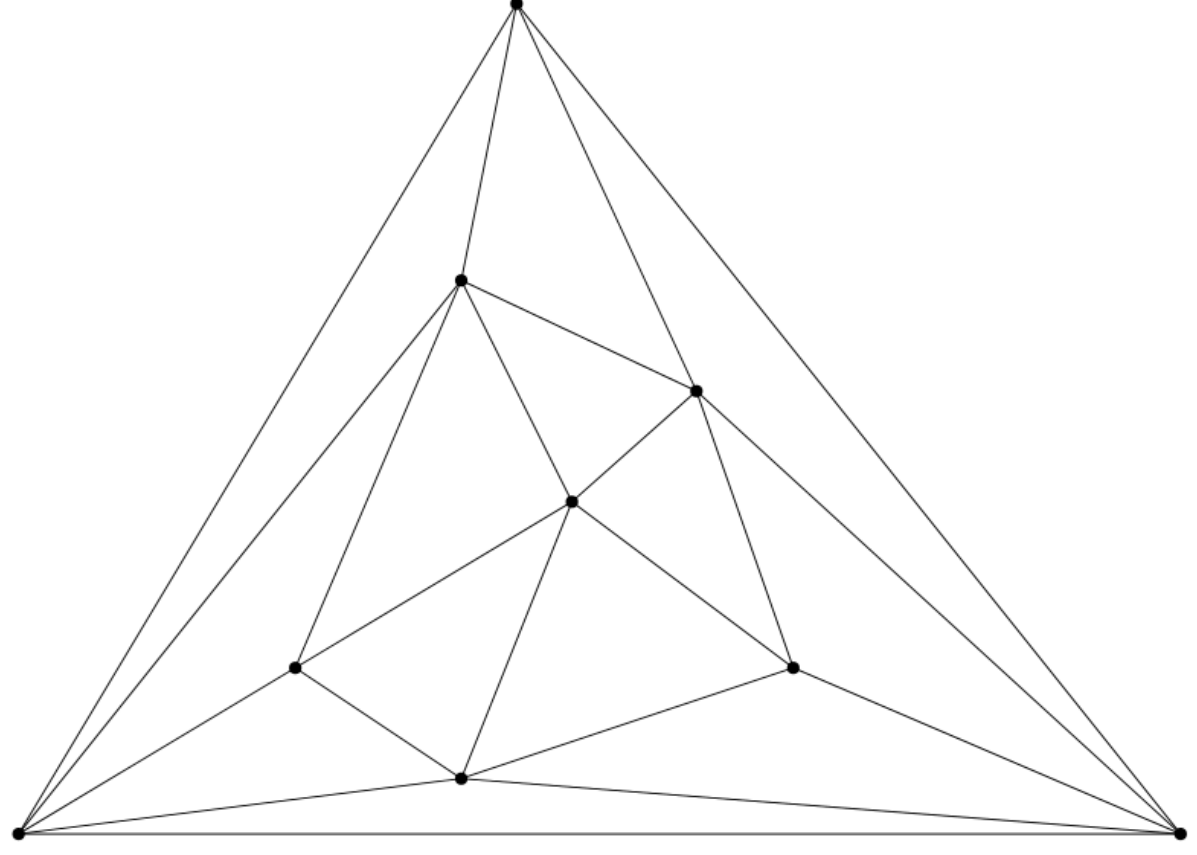
5 4 6 7 8 9 2

5 6 7 8 9 4 2

Correct

Correct

3.



1 / 1 point

Consider the triangulation in the image above that was the result of the randomized incremental approach. We wonder how many triangles were created in the last step of the incremental construction, including triangles created by doing edge flips. Recall that the outer three vertices were added in the beginning.

How many triangles were created at least (in the last step)?

How many triangles were created in expectation (in the last step)?

How many triangles were created at most (in the last step)?

At least 5, on average 6.333, at most 7.

At least 5, on average 8.333, at most 10.

At least 4, on average 4.667, at most 5.

Correct

Correct, a vertex of degree d created $2 \cdot d - 3$ triangles in total during its insertion.

https://www.coursera.org/learn/geometric-algorithms/quiz/2/en/randomized-incremental-construction/atamp?redirectToCover=true

1/1